

Integrated Power Supply for TFT-LCD

Features

- Input Supply Range: 2.5V ~ 5.0V
- I²C Interface
- Current Mode Sync. Boost Converter for AVDD
 - Programmable Switching Frequency
 - 4.0V to 7.0V Programmable Output
 - Driving ability 400mA
- Sync. Buck Converter for V_{CORE} and VIO
 - Programmable Switching Frequency
 - 0.8V to 2.06V Programmable Output for V_{CORE}
 - 1.0V to 2.55V Programmable Output for VIO
- Programmable SIBO Converter For V_{GH} / V_{GL}
- Inverting control for AVEE
 - Programmable Switching Frequency
 - -4.0V to -7.0V Programmable Output
 - Driving ability 400mA
- Low Dropout Regulator
 - 1.7V to 2.8V Programmable Output
- VCOM Operation Amplifier
 - ±200mA Shift Circuit Current
 - Programmable 7-Bit VCOM Calibrator
- Programmable 2-CH Gamma
 - ±20mA Shift Circuit Current
- Programmable Voltage Detector
- Protection Function
 - Over-Current Protection (OCP)
 - Over-Temperature Protection (OTP)
 - Over-Voltage Protection (OVP)
 - Output Under Voltage Protection (UVP)
 - Short Circuit Protection (SCP)
- Available in WQFN3.5X3.5-28 pin Package

General Description

The NVP2515 is an I2C interface programmable PMIC. The IC includes one synchronous boost converter for AVDD, a SIBO converter for VGH and VGL, two synchronous buck converter for V_{CORE} and VIO, one inverting converter for AVEE, one LDO regulator, one voltage detector, one programmable V_{COM} generator with a high-speed amplifier for DVCOM, two high-speed operation amplifiers for GMA, for active-matrix, thin-film transistor, liquid-crystal displays (LCDs).

PMIC is available in a 28-pin thin WQFN3.5X3.5 package.

Applications

- NB PCs
- Table PCs

Ordering Information

ORDER NUMBER	MARKING	TEMP. RANGE	PACKAGE (Green)
NVP2515UG1U	2515	-40°C to 85°C	WQFN3.5X3.5-28

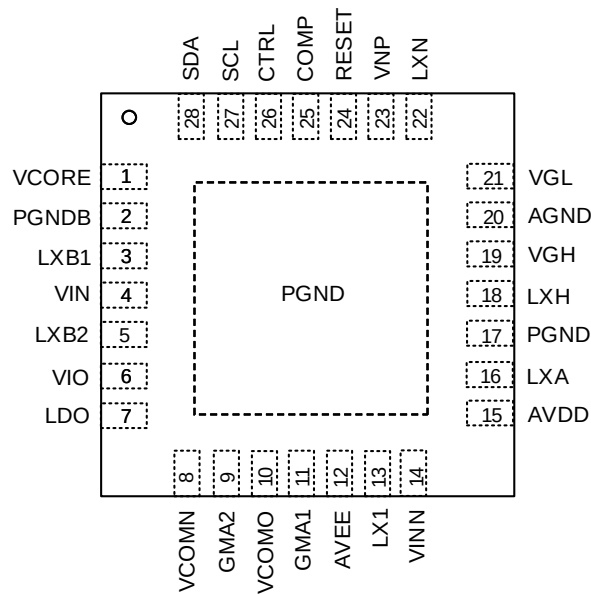
Note: UG: WQFN3.5X3.5-28

1: Bonding Code

U: Tape & Reel

Green : Lead Free / Halogen Free.

Pin Configuration



NVP2515 WQFN3.5X3.5-28

Note: Recommend connecting the Thermal Pad to the Ground for excellent power dissipation

Absolute Maximum Rating

VIN, VINP, VINN to PGND	0.3V to 10V
SDA, SCL to AGND	-0.3V to 7.0V
VIN, SDA, SCL to PGND (100mS pulse)	0.3V to 10V
VINP, VINN to PGND (100mS pulse)	0.3V to 10V
PGND, PGNDB to AGND	±0.3V
COMP, RESET, CTRL to AGND	-0.3V to 6.0V
AVDD, LXA, GMA1 to PGND	-0.3V to 7.0V
LXB1, LXB2 to PGNDB	-0.3V to 7.0V
VGH, LXH to PGND	-0.3V to 30V
VCORE, VIO, LDO to PGND	-0.3V to 7.0V
VCOMN, VCOM to PGND	-6.5V to 6.0V
AVEE, LXI, GMA2 to PGND	-0.3V to -7.0V
VGL, LXN to PGND	-0.3V to -20V

Thermal Resistance Junction to Ambient, (θ_{JA})	
WQFN3.5X3.5-28	31.2°C/W
Continuous Power Dissipation ($T_A = +25^\circ\text{C}$)	
WQFN3.5X3.5-28	3.2W
Thermal Resistance Junction to Case, (θ_{JC})	
WQFN3.5X3.5-28	2.2°C/W
Operating Temperature Range	-40°C to 85°C
Storage Temperature Range	-65°C to 150°C
Junction Temperature	160°C
Reflow Temperature (soldering, 10sec)	250°C
Electrostatic Discharge, VESD (note1)	
HBM	3kV/8kV
MM	300V
CDM	750V

Note1 :

Devices are ESD sensitive. Handling precaution is recommend.
Except SDA, SCL, VIN pin 8kV for HBM

Electrical Characteristics

($V_{IN}=V_{INN}=3.3V$, $AVDD=5V$, $AVEE=-5V$, $V_{CORE}=1.2V$, $VIO=1.8V$, $VLDO=2.5V$, $VGH=8.2V$, $VGL=-7V$, $VCOM=-1V$, $T_A = 25^{\circ}C$)

The device is not guaranteed to function outside its operating conditions. Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}C$, unless otherwise specified.

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
GENERAL						
VIN Input Voltage Range	VIN		2.5	---	5.0	V
VINN Input Voltage Range	VINN		2.5	---	5.0	V
VIN Quiescent Current	IVIN		---	---	5	mA
VIN Under-voltage Lockout	VUVLO	VIN rising, turn on IC	2.3	2.4	2.5	V
		Hysteresis	---	0.5	---	
Duration to Trigger Fault Condition	tFDT	Boost, Buck, Inverting, SIBO	---	60	---	ms
Thermal Shutdown Protection	TSd	Temperature rising	---	150	---	°C
	ΔTSd	Hysteresis	---	20	---	
AVDD Boost Converter						
Output Voltage Range	VAVDD	Programmable	4.0	---	7.0	V
Output Voltage Bit			---	6	---	Bits
Output Voltage Resolution			---	0.05	---	V
Output Voltage Accuracy		No load	-1	--	+1	%
Output Load Regulation		AVDD=5V, Iout=10 to 350mA, fSW_LXA=750kHz	---	100	---	mV/A
Output Line Regulation		VIN=2.5V to 5.0V	---	0.1	0.15	%/V
Switching Frequency	FSW_LXA	Programmable	600	--	1225	kHz
Switching Frequency Bit			---	3	---	Bits
Switching Frequency Accuracy		715kHz	-5	---	5	%
		Others	-10	---	10	
Maximum Duty Cycle	DMAX_AVDD		86	90	---	%
UVP Percentage	UVP_AVDD		---	80	---	%
LX Leakage Current	ILXA	VLX = 6.2V	---	1	5	μA
LX Current Limit	ILIM_AVDD	Programmable	0.5	---	2	A
Current Limit Bit			---	2	---	Bits
Current Limit Resolution			---	0.5	---	A
Output Current Ability		VIN=2.8V, AVDD=5V, L=10uH	400	---	---	mA
VCORE Buck Converter						
Output Voltage Range	VCORE	Programmable	0.8	---	2.06	V
Output Voltage Bit			---	6	---	Bits
Output Voltage Resolution			---	0.02	---	V
Output Voltage Accuracy		No load	-1	---	+1	%
Output Load Regulation		VCORE=1.2V, Iout=10 to 700mA,	---	50	---	mV/A
Output Line Regulation		VIN=2.5V to 5.0V	---	0.1	0.15	%/V
Switching Frequency	FSW_LXB1	Programmable	600	---	1225	kHz
Switching Frequency Bit			---	3	---	Bits
Switching Frequency Accuracy		715kHz	-5	---	5	%
		Others	-10	---	10	
Maximum Duty Cycle	DMAX_VCORE		---	100	---	%
UVP Percentage	UVP_VCORE		---	80	---	%
LXB1 Leakage Current			---	1	5	μA
LXB1 Current Limit	ILIM_CORE		---	1.5	---	A
Output Current Ability		VIN=3.3V, VCORE=1.2V fSW_LXB1=1225kHz	700	---	---	mA

Electrical Characteristics (Continued)

($V_{IN}=3.3V$, $AVDD=5V$, $AVEE=-5V$, $V_{core}=1.2V$, $V_{IO}=1.8V$, $VLDO=2.5V$, $V_{GH}=8.2V$, $V_{GL}=-7V$, $V_{COM}=-1V$, $T_A = 25^{\circ}C$)

The device is not guaranteed to function outside its operating conditions. Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}C$, unless otherwise specified.

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
VIO Buck Converter						
Output Voltage Range	V_{IO}	Programmable	1.0	---	2.55	V
Output Voltage Bit			---	5	---	Bits
Output Voltage Resolution			---	0.05	---	V
Output Voltage Accuracy		No load	-1	---	+1	%
Output Load Regulation		$V_{IO}=1.8V$, $I_{out}=10$ to $350mA$,	---	50	---	mV/A
Output Line Regulation		$V_{IN}=2.5V$ to $5.0V$	---	0.1	0.15	%/V
Switching Frequency	F_{SW_LXB2}	Programmable	600	---	1225	kHz
Switching Frequency Bit			---	3	---	Bits
Switching Frequency Accuracy		715kHz	-5	---	5	%
		Others	-10	---	10	
Maximum Duty Cycle	D_{MAX_VIO}		---	100	---	%
UVP Percentage	UVP_VIO		---	80	---	%
LX2 Leakage Current			---	1	5	μA
LXB2 Current Limit			---	1.3	---	A
Output Current Ability		$V_{IN}=3.3V$, $V_{IO}=1.8V$, $f_{SW_LXB2}=1225kHz$	300	---	---	mA
Inverting Converter (AVEE)						
Output Voltage Range	V_{AVEE}	Programmable	-7.0	---	-4.0	V
Output Voltage Bit			---	5	---	Bits
Output Voltage Resolution			---	0.1	---	V
Output Voltage Accuracy		No load	-1.5	---	+1.5	%
Output Load Regulation		$V_{AVEE}=-5V$, $I_{out}=10$ to $300mA$, $F_{SW_LXA}=715kHz$	---	200	---	mV/A
Output Line Regulation		$V_{IN}=2.5V$ to $5.0V$	---	0.1	0.15	%/V
Switching Frequency	F_{SW_LXI}		---	F_{SW_LXA}	---	kHz
Switching Frequency Accuracy		Compare with setting	-10	---	10	%
UVP Percentage	UVP_AVEE		---	70	---	%
LXI Current Limit			1	---	2.5	A
Current Bit				2		Bits
Current Limit Resolution			---	0.5	---	A
Output Current Ability		$V_{AVEE}=-5V$	400	---	---	mA
SIBO Converter - VGH						
Output Voltage Range	V_{GH}	0.2V/step, resolution = 0	6.0	---	12	V
		0.5V/step, resolution = 1	12.5		28	%
Output Voltage Bit			---	5	---	Bits
Output Voltage Accuracy		No load	-3	---	3	%
Switching Frequency	F_{SW_LXH}	Programmable	500	---	1430	kHz
Switching Frequency Bit			---	3	---	Bits
Switching Frequency Accuracy		715kHz	-5	---	5	%
		Others	-10	---	10	
UVP Percentage	UVP_VGH		---	80	---	%
LXH Current Limit			---	0.7	---	A
Output Current Ability		$V_{GH}=18V$, $f_{SW_LXH}=715kHz$	10	---	---	mA
LXP Leakage Current		$V_{LXP} = 30V$	---	1	20	μA

Electrical Characteristics (Continued)

($V_{IN}=3.3V$, $AVDD=5V$, $AVEE=-5V$, $V_{core}=1.2V$, $V_{IO}=1.8V$, $VLDO=2.5V$, $V_{GH}=8.2V$, $V_{GL}=-7V$, $V_{COM}=-1V$, $T_A = 25^{\circ}C$)

The device is not guaranteed to function outside its operating conditions. Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}C$, unless otherwise specified.

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
SIBO Converter - VGL						
Output Voltage Range	V_{GL}	Programmable	-18	---	-5.4	V
Output Voltage Bit			---	6	---	Bits
Output Voltage Resolution			---	0.2	---	V
Output Voltage Accuracy		No load	-3	---	3	%
Switching Frequency	F_{SW_LXN}	Programmable	---	F_{SW_LXN}	---	kHz
Switching Frequency Bit			---	3	---	Bits
Switching Frequency Accuracy		715kHz	-5	---	5	%
		Others	-10	---	10	
UVP Percentage	UVP_VGL		---	80	---	%
Current Limit	I_{GL_LIM}		---	700	---	mA
Output Current Ability		$V_{GL}=-12V$, $f_{OSC_VGL}=715kHz$	10	---	---	mA
LXN Leakage Current		$V_{LXN} = -18V$	---	1	20	μA
Low Drop Output Regulator (LDO)						
Output Voltage Range	V_{LDO}	Programmable	1.7	---	2.8	V
Output Voltage Bit			---	4	---	Bits
Output Voltage Resolution			---	0.1	---	V
Output Voltage Accuracy		No load	-1	---	1	%
Output Load Regulation		$I_{out}=1$ to 100mA,	-1	---	1	mV/A
Output Line Regulation		$V_{IN}=2.5V$ to 5.0V, $I_{out}=10mA$	---	0.1	0.15	%/V
Output Current Ability			100	---	---	mA
Operational Amplifiers (VCOM Amplifier)						
Input Bias Current			-100	---	+100	nA
Input Offset Voltage		Unit gain, $V_{OUT}=-1V$	-15	---	15	mV
Output Voltage Range	V_{COM}		-3.6	---	2.02	V
Output Voltage Bit			---	7	---	Bits
Output Voltage Resolution			---	0.01	---	V
Output Voltage Accuracy	$V_{acc-COM}$	No load	-15	---	+15	mV
Output Voltage Swing High	V_{OH}	NEG=AVEE, $I_{out}=100\mu A$	AVDD -0.03	AVDD -0.05	---	V
		NEG=AVEE, $I_{out}=75mA$	AVDD-1.9	AVDD-1.5	---	V
Output Voltage Swing Low	V_{OL}	NEG=AVDD, $I_{out}=100\mu A$	---	AVEE +0.05	AVEE +0.03	V
		NEG=AVDD, $I_{out}=75mA$	---	AVEE+1.5	AVEE+1.9	V
Output current Ability			---	± 75	---	mA
Short-Circuit Current			---	± 150	---	mA
Integral Nonlinearity Error	INL	Unit gain, $V_{OUT}=-2.56V$ to 1.06V	---	1	---	LSB
Differential Nonlinearity Error	DNL	Unit gain, $V_{OUT}=-2.56V$ to 1.06V	---	1	---	LSB
GMA1, GMA2						
GMA1 Output Voltage Range	V_{GMA1}		AVDD -1.26	---	AVDD	V
GMA2 Output Voltage Range	V_{GMA2}		AVEE	---	AVEE +1.26	V
Output Voltage Bit			---	6	---	Bits
Output Voltage Resolution			---	0.02	---	V
Output Voltage Accuracy	$V_{acc-COM}$	No load	-20	---	+20	mV
Short-Circuit Current			---	20	---	mA

Electrical Characteristics (Continued)

($V_{IN}=3.3V$, $AVDD=5V$, $AVEE=-5V$, $V_{core}=1.2V$, $V_{IO}=1.8V$, $VLDO=2.5V$, $V_{GH}=8.2V$, $V_{GL}=-7V$, $V_{COM}=-1V$, $T_A = 25^{\circ}C$)

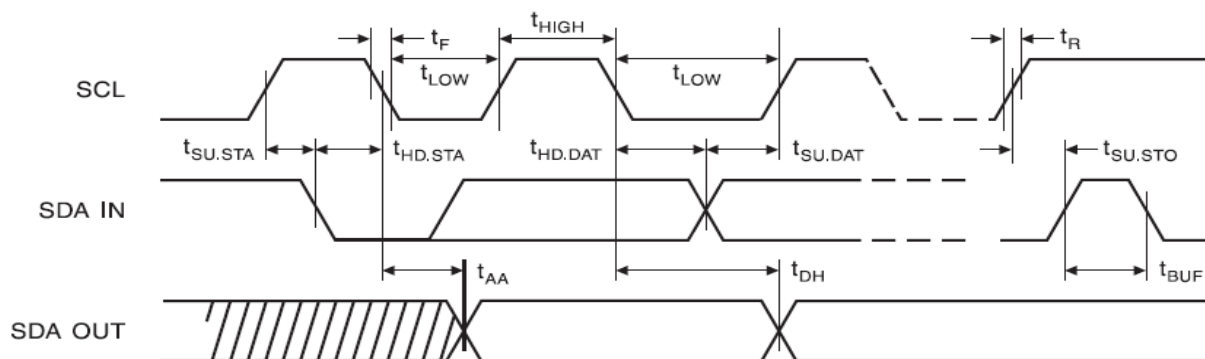
The device is not guaranteed to function outside its operating conditions. Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}C$, unless otherwise specified.

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Voltage Detector (RESET)						
Detection Voltage	V_{RESET}	0.1/step	2.0	--	2.7	V
Detection Voltage Bit			---	3	---	Bits
Detection Voltage Resolution			---	0.1	---	V
Detection Voltage Accuracy			-2	---	2	%
CTRL						
Threshold Voltage	V_{IH}		1.0	---	---	V
	V_{IL}		---	---	0.5	

I²C Timing Characteristics

PARAMETER	Symbol	CONDITIONS	MIN	TYP	MAX	UNITS
Logic Input High Voltage Threshold	V_{IH}	SDC, SCL	1.2	---	---	V
Logic Input Low Voltage Threshold	V_{IL}	SDC, SCL	---	---	0.4	V
SDA Output Sink Current	$I_{SINK-SDA}$	$V_{SDA}=0.4V$	---	6	---	mA
Serial-Clock Frequency	F_{SCL}		0.2	400	1000	kHz
Data Setup Time	$t_{SU,DAT}$		100	---	---	ns
Data Hold Time	$t_{HD,DAT}$		---	---	300	ns
SDA and SCL Rise Time	t_R		$20+0.1CB$	---	200	ns
SDA and SCL Fall Time	t_F		$20+0.1CB$	---	200	ns
SDA and SCL Input Capacitance			---	5	---	pF
START Condition of Setup Time	$T_{SU,STA}$		0.25	---	---	μs
START Condition of Hold Time	$T_{HD,STA}$	10% of SDA to 90% of SCL	0.25	---	---	μs
Bus Free Time Between STOP and START Conditions	t_{BUF}		0.5	---	---	μs
SDA_ACK on Voltage			---	---	0.4	V
Pulse Width of Suppressed Spike	T_{SP}		---	---	50	ns
Clock Pulse Width Low	t_{LOW}		1.3	---	---	μs
Clock Pulse Width High	t_{HIGH}		0.6	---	---	μs
Bus Free Time Between Stop	$t_{SU,STO}$		0.6	---	---	μs
Clock Low to Data Out Valid	t_{AA}		0.1	---	0.9	μs
Data Out Hold Time	t_{DH}		50	---	---	ns

I²C Bus Timing

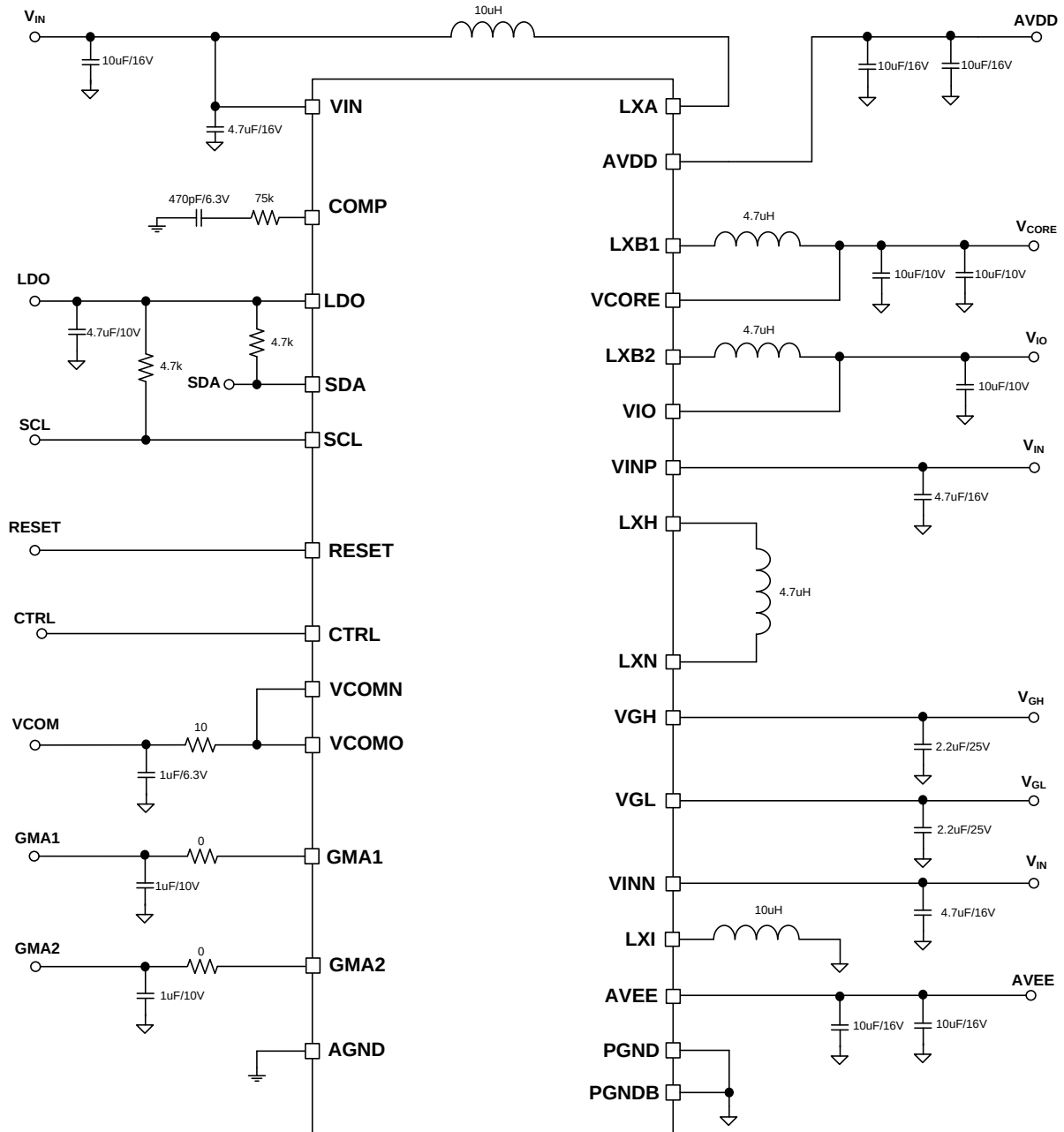


**Pin Description**

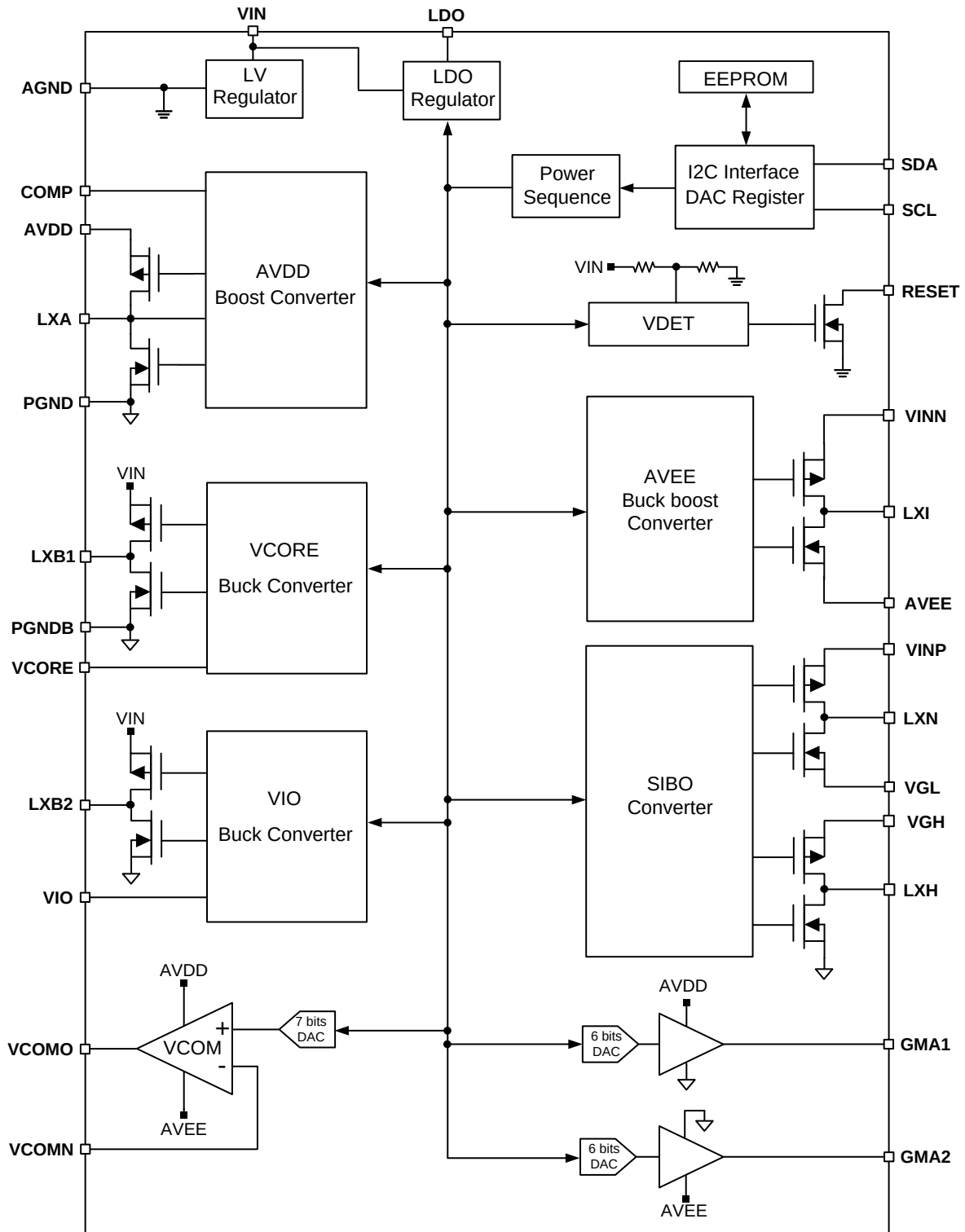
PIN	Symbol	I/O/P	FUNCTION
1	VCORE	I	VCORE output feedback.
2	PGNDB	G	Power ground of Buck converter.
3	LXB1	P	VCORE buck switching node.
4	VIN	P	Power supply voltage input of IC.
5	LXB2	P	VIO buck switching node.
6	VIO	I	VIO output feedback.
7	LDO	O	LDO Output
8	VCOMN	I	Inverted-terminal of VCOM buffer.
9	GMA2	O	Gamma2 output pin.
10	VCOMO	O	VCOM OP-amp output.
11	GMA1	O	Gamma1 output pin.
12	AVEE	O	Output of AVEE Invert Converter.
13	LXI	P	Negative AVEE switching node.
14	VINN	P	Power supply voltage input of AVEE.
15	AVDD	O	Output of AVDD Boost Converter
16	LXA	P	Switching Pin of AVDD Boost Converter
17	PGND	G	Power ground
18	LXH	O	VGH switching pin of SIBO Converter.
19	VGH	O	VGH SIBO converter output voltage.
20	AGND	P	Analog Ground.
21	VGL	O	VGL SIBO converter output voltage
22	LXN	O	VGL switching pin of SIBO Converter.
23	VINP	P	Power supply voltage input SIBO converter.
24	RESET	O	Output of voltage detector function.
25	COMP	O	AVDD Boost converter compensation input.
26	CTRL	I	Control pin of Stand-by function.
27	SCL	I	Clock Input for I2C interface.
28	SDA	I/O	Serial data input/output for I2C Interface.
29	PGND (Exposed Pad)	P	GND. The exposed pad must be soldered to a large PCB and connected to GND for maximum power dissipation.

Note: P: power/ground; I: Input; O: Output; I/O: Bi-direction

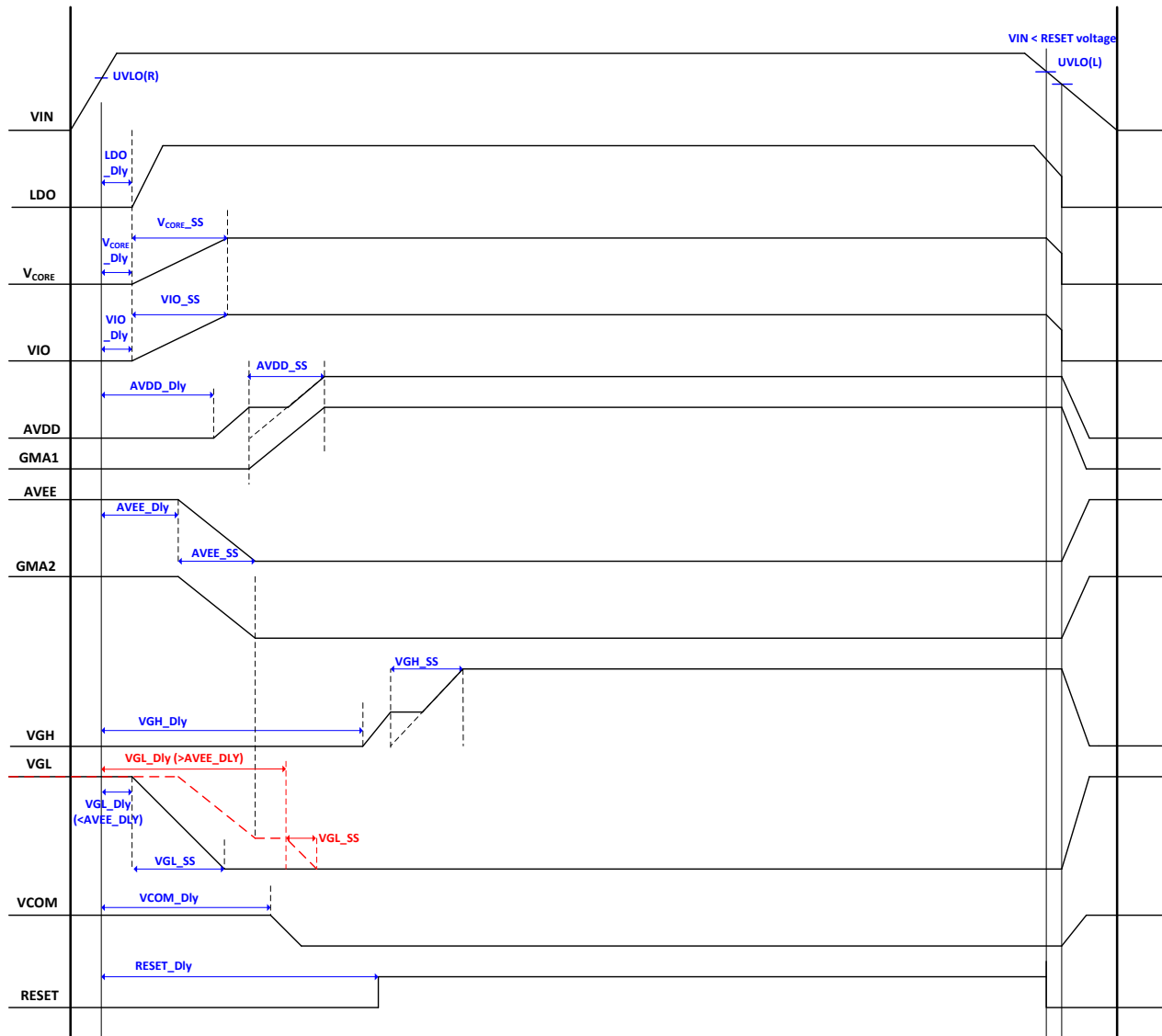
Application Circuit



Block Diagram



Timing Diagram



Note :

1. VGL_SS should be set to equal to AVEE_SS.
2. Vcom power on delay time range: ~155ms, Power off Vcom to GND $\leq$ 1ms.
3. The voltage of VGL must be lower than or equal to AVEE .



Device Address Setting

PMIC I²C Command

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
PMIC Slave Address 7'b0100011							Read:1 Write:0

Write Operation

(a) Write Single Byte to DAC Register

Example : Writing 29h to DAC Address 07h

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	0	1	0	0	0	1	1	0	SLAVE ACK	0	0	0	0	0	1	1	1	SLAVE ACK	0	0	1	0	1	0	0	1	SLAVE ACK	Stop

(b) Write Multi Byte to DAC Register

Example : Writing 29h, 2Ah, 2Bh to DAC Address 06h, 07h, 08h

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	0	1	0	0	0	1	1	0	SLAVE ACK	0	0	0	0	0	1	1	0	SLAVE ACK	0	0	1	0	1	0	0	1	SLAVE ACK	

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
0	0	1	0	1	0	1	0	SLAVE ACK	0	0	1	0	1	0	1	1	SLAVE ACK	Stop

(C) Write All DAC Register into EEPROM

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
0	1	0	0	0	1	1	0	SLAVE ACK	1	1	1	1	1	1	1	1	SLAVE ACK	1	0	0	0	0	0	0	0	SLAVE ACK	Stop

Read Operation

(a) Read data form DAC register

Example : Reading Data form DAC Register Address 08h, 09h, 0Ah, 0Bh

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	0	1	0	0	0	1	1	0	SLAVE ACK	1	1	1	1	1	1	1	1	SLAVE ACK	0	0	0	0	0	0	0	0	SLAVE ACK	Stop

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	0	1	0	0	0	1	1	0	SLAVE ACK	0	0	0	0	1	0	0	0	SLAVE ACK	

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	
Repeat Start	0	1	0	0	0	1	1	1	SLAVE ACK	D	D	D	D	D	D	D	D	Master ACK	D	D	D	D	D	D	D	D	Master ACK

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
D	D	D	D	D	D	D	D	Master ACK	D	D	D	D	D	D	D	D	Master N-ACK	Stop

(b) Read data form EEPROM

Example : Reading Data form EEPROM Address 03h, 04h, 05h, 06h

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	0	1	0	0	0	1	1	0	SLAVE ACK	1	1	1	1	1	1	1	1	SLAVE ACK	0	0	0	0	0	0	0	1	SLAVE ACK	Stop

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	0	1	0	0	0	1	1	0	SLAVE ACK	0	0	0	0	0	0	1	1	SLAVE ACK	

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	
Repeat Start	0	1	0	0	0	1	1	1	SLAVE ACK	D	D	D	D	D	D	D	D	Master ACK	D	D	D	D	D	D	D	D	Master ACK

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
D	D	D	D	D	D	D	D	Master ACK	D	D	D	D	D	D	D	D	Master N-ACK	Stop

D-VCOM I²C Command

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Configuration Parameter VCOM Slave Address 7'b1001111							Read:1 Write:0

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
VCOM data [7:1]							CMD

Write Operation

(a) Write Single Byte to DAC Register

Example : Writing 77h (7bit data) to DAC

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	1	0	0	1	1	1	1	0	SLAVE ACK	1	1	1	0	1	1	1	1	SLAVE ACK	Stop

b) Write Single Byte to DAC Register and EEPROM

Example : Writing 77h (7bit data) to DAC and EEPROM

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	1	0	0	1	1	1	1	0	SLAVE ACK	1	1	1	0	1	1	1	0	SLAVE ACK	Stop

Read Operation

(a) Read Single data from DAC when DAC data same as EEPROM

Example : Read Data from DAC

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	1	0	0	1	1	1	1	1	SLAVE ACK	D	D	D	D	D	D	D	1	SLAVE ACK	Stop

(b) Read Single data from DAC when DAC data different to EEPROM

Example : Read Data from DAC

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0		
Start	1	0	0	1	1	1	1	1	SLAVE ACK	D	D	D	D	D	D	D	0	SLAVE ACK	Stop



Register Map

Inst/Prar	7	6	5	4	3	2	1	0	register default	description
00h	EN_VCOM	EN_LDO	EN_VCORE	EN_VIO	EN_VGL	EN_VGH	EN_AVEE	EN_AVDD	FFh	Channel setting [7:0] : 0h=Disable 1h=Enable
01h	EN_High Resolution	EN_VIO_PWM	EN_VCORE_PWM	PRE_AVDD	EN_CTRL	EN_RESET	EN_GMA2	EN_GMA1	07h	Channel setting [7:0] : 0h=Disable 1h=Enable
02h	EN_VCOM Discharge	EN_LDO Discharge	EN_VIO Discharge	EN_Vcore Discharge	EN_VGL Discharge	EN_VGH Discharge	EN_AVEE Discharge	EN_AVDD Discharge	00h	Channel discharge setting [7:0] : 0h=Disable 1h=Enable
03h									14h	AVDD Voltage[5:0] : adjust the AVDD output, range : 4.0V ~ 7.0V Resolution : 0.05V/step Default : 5.0V
04h									05h	AVEE Voltage[4:0] : adjust the AVEE output, range : -4.0V ~ -7.0V Resolution : 0.1V/step Default : -4.5V
05h									08h	VGH Voltage[4:0] : adjust VGH, range base on EN_High Reslution bit 0x01h[7] 0x01h [7] : 0 VGH range : 6V ~ 12V, Resolution : 0.2V/step 0x01h [7] : 1 VGH range 12.5V ~ 28V Resolution : 0.5V/step
06h									08h	VGL Voltage [5:0] : adjust VGL output, range : -5.4V ~ -18.0V, Resolution : 0.2V/step Default : -7.0V
07h									14h	Vcore Voltage [5:0] : adjust Vcore output, range : 0.8V ~ 2.06V, Resolution : 0.02V/step Default : 1.2V
08h									10h	VIO Voltage [4:0] : adjust VIO output, range : 1.0V ~ 2.55V, Resolution : 0.05V/step Default : 1.8V
09h									08h	VLDO Voltage [3:0] : adjust VLDO output, range 1.7V ~ 2.8V, Resolution : 0.1V/step Default : 2.5V
0Ah									80h	VCOM Voltage [7:1] : adjust VCOM output, range : VCOM_MIN ~ VCOM_MIN+1.27V, Resolution : 0.01V/step Default : VCOM_MIN+0.64V
0Bh									02h	RESET Voltage [2:0] : adjust RESET voltage, range 2.0V~ 2.7V, Resolution : 0.1V/step Default : 2.2V (default)
0Ch									0Ah	GMA1 voltage [5:0] : adjust GMA1 output, range AVDD ~ AVDD-1.26V Resolution : 0.02V/step Default : AVDD - 0.2V
0Dh									0Ah	GMA2 voltage [5:0] : adjust GMA2 output, range AVEE ~ AVEE+1.26V Resolution : 0.02V/step Default : AVEE+0.2V
0Eh	AVDD Current limit [7:6]		AVDD LX_Slew Rate [5:3]						61h	AVDD LX_FREQ. [2:0] : adjust PAVDD LX_FREQ., range : 600kHz ~ 1225kHz, 0h : 600kHz 1h : 715kHz(default) 2h : 800kHz 3h : 933kHz 4h : 1000kHz 5h : 1225kHz 6h : 1225kHz 7h : 1430kHz AVDD LX_Slew Rate [5:3] : adjust AVDD LX_Slew Rate., 0h ~ 07h : Level0 ~ Level7 Default : Level2 AVDD Current limit [7:6] : adjust PAVDD Current Limit., 0h : 0.5A 1h : 1A(default) 2h : 1.5A 3h : 2.0A
0Fh			AVDD_SS Time [6:4]						12h	AVDD_Delay Time [3:0] : adjust AVDD_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 10ms AVDD_SS Time[6:4] : adjust AVDD_SS Time, range : 2ms ~ 16ms Resolution : 2ms/step Default : 4ms
10h			AVEE_SS Time [6:4]						92h	AVEE_Delay Time [3:0] : adjust AVEE_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 10ms AVEE_SS Time[6:4] : adjust AVEE_SS Time, range : 4ms ~ 16ms Resolution : 2ms/step Default : 4ms
11h									12h	LXHLXN_FREQ. [2:0] : adjust LXHLXN_FREQ., range 500kHz ~ 1430kHz, 0h : 500kHz 1h : 600kHz 2h : 715kHz(default) 3h : 800kHz 4h : 933kHz 5h : 1000kHz 6h : 1225kHz 7h : 1430kHz LXHLXN_Slew Rate [4:3] : adjust LXHLXN_Slew Rate., 0h : Fastest 1h : Fast 2h : Normal(default) 3h : Slow
12h									05h	VGH_Delay Time [3:0] : adjust VGH_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 25ms VGH_SS Time[6:4] : adjust VGH_SS Time, range : 4ms ~ 12ms 0h : 4ms(default) 1h : 6ms 2h : 8ms 3h : 12ms
13h									03h	VGL_Delay Time [3:0] : adjust VGL_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 15ms VGL_SS Time[6:4] : adjust VGL_SS Time, range : 4ms ~ 16ms 0h : 4ms(default) 1h : 4ms 2h : 6ms 3h : 8ms 4h : 10ms 5h : 12ms 6h : 14ms 7h : 16ms
14h									15h	LXB1_FREQ. [2:0] : adjust LXB1_FREQ., range : 600kHz ~ 1225kHz, 0h : 600kHz 1h : 715kHz 2h : 800kHz 3h : 933kHz 4h : 1000kHz 5h : 1225kHz(default) LXB1_Slew Rate [4:3] : adjust LXB1_Slew Rate., 0h : Fastest 1h : Fast 2h : Normal(default) 3h : Slow
15h									11h	Vcore_Delay Time [3:0] : adjust Vcore_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 5ms Vcore_SS Time[5:4] : adjust Vcore_SS Time, range : 2ms ~ 8ms 0h : 2ms 1h : 4ms(default) 2h : 6ms 3h : 8ms
16h									15h	LXB2_FREQ. [2:0] : adjust LXB2_FREQ., range : 600kHz ~ 1225kHz, 0h : 600kHz 1h : 715kHz 2h : 800kHz 3h : 933kHz 4h : 1000kHz 5h : 1225kHz(default) LXB2_Slew Rate [4:3] : adjust LXB2_Slew Rate., 0h : Fastest 1h : Fast 2h : Normal(default) 3h : Slow
17h									12h	VIO_Delay Time [3:0] : adjust VIO_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 10ms VIO_SS Time[5:4] : adjust VIO_SS Time, range : 2ms ~ 8ms 0h : 2ms 1h : 4ms(default) 2h : 6ms 3h : 8ms
18h									03h	RESET_Delay Time [3:0] : adjust RESET_Delay Time, range : 0ms ~ 75ms Resolution : 5ms/step Default : 15ms
19h									02h	LDO_Delay Time [3:0] : adjust LDO_Delay Time, range : 0ms ~ 45ms Resolution : 5ms/step Default : 10ms
1Ah									05h	VCOM_Delay Time [4:0] : adjust VCOM_Delay Time, range : 0ms ~ 155ms Resolution : 5ms/step Default : 25ms VCOM Power off selection [5] : VCOM Power off behavior 0h : Follow UVLO_F(default) 1h : Follow RESET
1Bh									48h	LXI Current Limit [4:3] : adjust LXI Current Limit, range : 2.5A ~ 1.0A 0h : 2.5A 1h : 2.0A(default) 2h : 1.5A 3h : 1.0A LXI_Slew Rate [4:3] : adjust LXI_Slew Rate., 0h : Fastest 1h : Fast 2h : Normal(default) 3h : Slow
1Ch									16h	VCOM_MIN [4:0] : adjust VCOM_MIN Voltage, range : -3.6V ~ -0.75V
FEh									xxh	Write 0x0Ah commands to internal EEPROM : 0 : No Action 1 : Write DAC into EEPROM
FFh									xxh	00h : Read data from DAC 01h : Read data from EEPROM 80h : Write all DAC into EEPROM

Registers and DAC Settings

AVDD Voltage Register – Address 03h [5:0]

DAC Value	AVDD Voltage (V)	DAC Value	AVDD Voltage (V)	DAC Value	AVDD Voltage (V)	DAC Value	AVDD Voltage (V)
00h	4.0	10h	4.8	20h	5.6	30h	6.4
01h	4.05	11h	4.85	21h	5.65	31h	6.45
02h	4.1	12h	4.9	22h	5.7	32h	6.5
03h	4.15	13h	4.95	23h	5.75	33h	6.55
04h	4.2	14h	5.0	24h	5.8	34h	6.6
05h	4.25	15h	5.05	25h	5.85	35h	6.65
06h	4.3	16h	5.1	26h	5.9	36h	6.7
07h	4.35	17h	5.15	27h	5.95	37h	6.75
08h	4.4	18h	5.2	28h	6.0	38h	6.8
09h	4.45	19h	5.25	29h	6.05	39h	6.85
0Ah	4.5	1Ah	5.3	2Ah	6.1	3Ah	6.9
0Bh	4.55	1Bh	5.35	2Bh	6.15	3Bh	6.95
0Ch	4.6	1Ch	5.4	2Ch	6.2	3Ch	7.0
0Dh	4.65	1Dh	5.45	2Dh	6.25	3Dh	7.0
0Eh	4.7	1Eh	5.5	2Eh	6.3	3Eh	7.0
0Fh	4.75	1Fh	5.55	2Fh	6.35	3Fh	7.0

AVDD LXA Frequency Register – Address 0Eh [2:0]

DAC Value	LXA FREQ (KHz)	DAC Value	LXA FREQ (KHz)	DAC Value	LXA FREQ (KHz)	DAC Value	LXA FREQ (KHz)
00h	600	02h	800	04h	1000	06h	1225
01h	715	03h	933	05h	1225	07h	1225

AVDD LXA Slew Rate – Address 0Eh [4:3]

DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)
00h	Level 0 (Fastest)	02h	Level 2	04h	Level 4 (Normal)	06h	Level 6
01h	Level 1	03h	Level 3	05h	Level 5	07h	Level 7

AVDD LXA Current Limit Register – Address 0Eh [6:5]

DAC Value	Current Limit (A)	DAC Value	Current Limit (A)	DAC Value	Current Limit (A)	DAC Value	Current Limit (A)
00h	0.5	01h	1	02h	1.5	03h	2

AVDD Delay Time Register – Address 0Fh [3:0]

DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

AVDD Soft-Start Time Register – Address 0Fh [6:4]

DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)
00h	2	02h	6	04h	10	06h	14
01h	4	03h	8	05h	12	07h	16

AVEE Voltage Register – Address 04h [4:0]

DAC Value	AVEE Voltage (V)	DAC Value	AVEE Voltage (V)	DAC Value	AVEE Voltage (V)	DAC Value	AVEE Voltage (V)
00h	-4.0	08h	-4.8	10h	-5.6	18h	-6.4
01h	-4.1	09h	-4.9	11h	-5.7	19h	-6.5
02h	-4.2	0Ah	-5.0	12h	-5.8	1Ah	-6.6
03h	-4.3	0Bh	-5.1	13h	-5.9	1Bh	-6.7
04h	-4.4	0Ch	-5.2	14h	-6.0	1Ch	-6.8
05h	-4.5	0Dh	-5.3	15h	-6.1	1Dh	-6.9
06h	-4.6	0Eh	-5.4	16h	-6.2	1Eh	-7.0
07h	-4.7	0Fh	-5.5	17h	-6.3	1Fh	-7.0

AVEE Delay Time Register – Address 10h [3:0]

DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

AVEE Soft-start Time Register – Address 10h [6:4]

DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)
00h	4	02h	6	04h	10	06h	14
01h	4	03h	8	05h	12	07h	16

AVEE LXI Current Limit Register – Address 1Bh [4:3]

DAC Value	Current Limit (A)	DAC Value	Current Limit (A)	DAC Value	Current Limit (A)	DAC Value	Current Limit (A)
00h	2.5	01h	2.0	02h	1.5	03h	1

AVEE LXI Slew Rate Register – Address 1Bh [6:5]

DAC Value	Slew Rate (us/V)	DAC Value	Slew Rate (us/V)	DAC Value	Slew Rate (us/V)	DAC Value	Slew Rate (us/V)
00h	Fastest	01h	Fast	02h	Normal	03h	Slow

VGH Voltage Register (VGH_High_Resolution=0) – Address 05h [4:0]

DAC Value	VGH Voltage (V)	DAC Value	VGH Voltage (V)	DAC Value	VGH Voltage (V)	DAC Value	VGH Voltage (V)
00h	6.0	08h	7.6	10h	9.2	18h	10.8
01h	6.2	09h	7.8	11h	9.4	19h	11
02h	6.4	0Ah	8.0	12h	9.6	1Ah	11.2
03h	6.6	0Bh	8.2	13h	9.8	1Bh	11.4
04h	6.8	0Ch	8.4	14h	10	1Ch	11.6
05h	7.0	0Dh	8.6	15h	10.2	1Dh	11.8
06h	7.2	0Eh	8.8	16h	10.4	1Eh	12
07h	7.4	0Fh	9.0	17h	10.6	1Fh	12

VGH Voltage Register (VGH_High_Resolution=1) – Address 05h [4:0]

DAC Value	VGH Voltage (V)	DAC Value	VGH Voltage (V)	DAC Value	VGH Voltage (V)	DAC Value	VGH Voltage (V)
00h	12.5	08h	16.5	10h	20.5	18h	24.5
01h	13	09h	17	11h	21	19h	25
02h	13.5	0Ah	17.5	12h	21.5	1Ah	25.5
03h	14	0Bh	18	13h	22	1Bh	26
04h	14.5	0Ch	18.5	14h	22.5	1Ch	26.5
05h	15	0Dh	19	15h	23	1Dh	27
06h	15.5	0Eh	19.5	16h	23.5	1Eh	27.5
07h	16	0Fh	20	17h	24	1Fh	28

LXH/LXN Frequency Register – Address 11h [2:0]

DAC Value	LXP/LXN FREQ (KHz)	DAC Value	LXP/LXN FREQ (KHz)	DAC Value	LXP/LXN FREQ (KHz)	DAC Value	LXP/LXN FREQ (KHz)
00h	500	02h	715	04h	933	06h	1225
01h	600	03h	800	05h	1000	07h	1430

LXH/LXN Slew Rate Register – Address 11h [4:3]

DAC Value	Slew Rate (us/V)	DAC Value	Slew Rate (us/V)	DAC Value	Slew Rate (us/V)	DAC Value	Slew Rate (us/V)
00h	Fastest	01h	Fast	02h	Normal	03h	Slow

VGH Delay Time Register – Address 12h [3:0]

DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

VGH Soft-Start Time Register – Address 12h [5:4]

DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)
00h	4	01h	6	02h	8	03h	12

VGL Voltage Register – Address 06h [5:0]

DAC Value	VGL Voltage (V)	DAC Value	VGL Voltage (V)	DAC Value	VGL Voltage (V)	DAC Value	VGL Voltage (V)
00h	-5.4	10h	-8.6	20h	-11.8	30h	-15
01h	-5.6	11h	-8.8	21h	-12	31h	-15.2
02h	-5.8	12h	-9.0	22h	-12.2	32h	-15.4
03h	-6.0	13h	-9.2	23h	-12.4	33h	-15.6
04h	-6.2	14h	-9.4	24h	-12.6	34h	-15.8
05h	-6.4	15h	-9.6	25h	-12.8	35h	-16
06h	-6.6	16h	-9.8	26h	-13	36h	-16.2
07h	-6.8	17h	-10	27h	-13.2	37h	-16.4
08h	-7.0	18h	-10.2	28h	-13.4	38h	-16.6
09h	-7.2	19h	-10.4	29h	-13.6	39h	-16.8
0Ah	-7.4	1Ah	-10.6	2Ah	-13.8	3Ah	-17
0Bh	-7.6	1Bh	-10.8	2Bh	-14	3Bh	-17.2
0Ch	-7.8	1Ch	-11	2Ch	-14.2	3Ch	-17.4
0Dh	-8.0	1Dh	-11.2	2Dh	-14.4	3Dh	-17.6
0Eh	-8.2	1Eh	-11.4	2Eh	-14.6	3Eh	-17.8
0Fh	-8.4	1Fh	-11.6	2Fh	-14.8	3Fh	-18

VGL Delay Time Register – Address 13h [3:0]

DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

VGL Soft-Start Time Register – Address 13h [6:4]

DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)
00h	4	02h	6	04h	10	06h	14
01h	4	03h	8	05h	12	07h	16

VCORE Voltage Register – Address 07h [5:0]

DAC Value	VCORE Voltage (V)	DAC Value	VCORE Voltage (V)	DAC Value	VCORE Voltage (V)	DAC Value	VCORE Voltage (V)
00h	0.8	10h	1.12	20h	1.44	30h	1.76
01h	0.82	11h	1.14	21h	1.46	31h	1.78
02h	0.84	12h	1.16	22h	1.48	32h	1.8
03h	0.86	13h	1.18	23h	1.5	33h	1.82
04h	0.88	14h	1.2	24h	1.52	34h	1.84
05h	0.9	15h	1.22	25h	1.54	35h	1.86
06h	0.92	16h	1.24	26h	1.56	36h	1.88
07h	0.94	17h	1.26	27h	1.58	37h	1.9
08h	0.96	18h	1.28	28h	1.6	38h	1.92
09h	0.98	19h	1.3	29h	1.62	39h	1.94
0Ah	1.0	1Ah	1.32	2Ah	1.64	3Ah	1.96
0Bh	1.02	1Bh	1.34	2Bh	1.66	3Bh	1.98
0Ch	1.04	1Ch	1.36	2Ch	1.68	3Ch	2.0
0Dh	1.06	1Dh	1.38	2Dh	1.7	3Dh	2.02
0Eh	1.08	1Eh	1.4	2Eh	1.72	3Eh	2.04
0Fh	1.1	1Fh	1.42	2Fh	1.74	3Fh	2.06

LXB1 Frequency Register – Address 14h [2:0]

DAC Value	LXB1 FREQ (KHz)	DAC Value	LXB1 FREQ (KHz)	DAC Value	LXB1 FREQ (KHz)	DAC Value	LXB1 FREQ (KHz)
00h	600	02h	800	04h	1000	06h	1225
01h	715	03h	933	05h	1225	07h	1225

LXB1 Slew Rate – Address 14h [4:3]

DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)
00h	Fastest	01h	Fast	02h	Normal	03h	Slow

VCORE Delay Time Register – Address 15h [3:0]

DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

VCORE Soft-Start Time Register – Address 15h [5:4]

DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)
00h	2	01h	4	02h	6	03h	8

VIO Voltage Register – Address 08h [4:0]

DAC Value	VIO Voltage (V)	DAC Value	VIO Voltage (V)	DAC Value	VIO Voltage (V)	DAC Value	VIO Voltage (V)
00h	1.0	08h	1.4	10h	1.8	18h	2.2
01h	1.05	09h	1.45	11h	1.85	19h	2.25
02h	1.1	0Ah	1.5	12h	1.9	1Ah	2.3
03h	1.15	0Bh	1.55	13h	1.95	1Bh	2.35
04h	1.2	0Ch	1.6	14h	2.0	1Ch	2.4
05h	1.25	0Dh	1.65	15h	2.05	1Dh	2.45
06h	1.3	0Eh	1.7	16h	2.1	1Eh	2.5
07h	1.35	0Fh	1.75	17h	2.15	1Fh	2.55

LXB2 Frequency Register – Address 16h [2:0]

DAC Value	LXB2 FREQ (KHz)	DAC Value	LXB2 FREQ (KHz)	DAC Value	LXB2 FREQ (KHz)	DAC Value	LXB2 FREQ (KHz)
00h	600	02h	800	04h	1000	06h	1225
01h	715	03h	933	05h	1225	07h	1225

LXB2 Slew Rate – Address 16h [4:3]

DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)	DAC Value	S.R. Time (ms)
00h	Fastest	01h	Fast	02h	Normal	03h	Slow

VIO Delay Time Register – Address 17h [3:0]

DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

VIO Soft-Start Time Register – Address 17h [5:4]

DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)	DAC Value	SS Time (ms)
00h	2	01h	4	02h	6	03h	8

LDO Voltage Register – Address 09h [3:0]

DAC Value	LDO Voltage (V)	DAC Value	LDO Voltage (V)	DAC Value	LDO Voltage (V)	DAC Value	LDO Voltage (V)
00h	1.7	04h	2.1	08h	2.5	0Ch	2.8
01h	1.8	05h	2.2	09h	2.6	0Dh	2.8
02h	1.9	06h	2.3	0Ah	2.7	0Eh	2.8
03h	2.0	07h	2.4	0Bh	2.8	0Fh	2.8

LDO Delay Time Register – Address 19h [3:0]

DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)	DAC Value	Delay Time (mS)
00h	0	04h	20	08h	40	0Ch	45
01h	5	05h	25	09h	45	0Dh	45
02h	10	06h	30	0Ah	45	0Eh	45
03h	15	07h	35	0Bh	45	0Fh	45

RESET Voltage Register – Address 0Bh [2:0]

DAC Value	RESET Voltage(V)	DAC Value	RESET Voltage(V)	DAC Value	RESET Voltage(V)	DAC Value	RESET Voltage(V)
00h	2.0	02h	2.2	04h	2.4	06h	2.6
01h	2.1	03h	2.3	05h	2.5	07h	2.7

RESET Delay Time Register – Address 18h [3:0]

DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)
00h	0	04h	20	08h	40	0Ch	60
01h	5	05h	25	09h	45	0Dh	65
02h	10	06h	30	0Ah	50	0Eh	70
03h	15	07h	35	0Bh	55	0Fh	75



VCOM Voltage Register – Address 0Ah [7:1]

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
VCOM Voltage [7:1]							--

DAC Value	VCOM Voltage(V)	DAC Value	VCOM Voltage(V)	DAC Value	VCOM Voltage(V)	DAC Value	VCOM Voltage(V)
00h	VCOM_MIN+0	20h	VCOM_MIN+0.32	40h	VCOM_MIN+0.64	60h	VCOM_MIN+0.96
01h	VCOM_MIN+0.01	21h	VCOM_MIN+0.33	41h	VCOM_MIN+0.65	61h	VCOM_MIN+0.97
02h	VCOM_MIN+0.02	22h	VCOM_MIN+0.34	42h	VCOM_MIN+0.66	62h	VCOM_MIN+0.98
03h	VCOM_MIN+0.03	23h	VCOM_MIN+0.35	43h	VCOM_MIN+0.67	63h	VCOM_MIN+0.99
04h	VCOM_MIN+0.04	24h	VCOM_MIN+0.36	44h	VCOM_MIN+0.68	64h	VCOM_MIN+1
05h	VCOM_MIN+0.05	25h	VCOM_MIN+0.37	45h	VCOM_MIN+0.69	65h	VCOM_MIN+1.01
06h	VCOM_MIN+0.06	26h	VCOM_MIN+0.38	46h	VCOM_MIN+0.70	66h	VCOM_MIN+1.02
07h	VCOM_MIN+0.07	27h	VCOM_MIN+0.39	47h	VCOM_MIN+0.71	67h	VCOM_MIN+1.03
08h	VCOM_MIN+0.08	28h	VCOM_MIN+0.40	48h	VCOM_MIN+0.72	68h	VCOM_MIN+1.04
09h	VCOM_MIN+0.09	29h	VCOM_MIN+0.41	49h	VCOM_MIN+0.73	69h	VCOM_MIN+1.05
0Ah	VCOM_MIN+0.10	2Ah	VCOM_MIN+0.42	4Ah	VCOM_MIN+0.74	6Ah	VCOM_MIN+1.06
0Bh	VCOM_MIN+0.11	2Bh	VCOM_MIN+0.43	4Bh	VCOM_MIN+0.75	6Bh	VCOM_MIN+1.07
0Ch	VCOM_MIN+0.12	2Ch	VCOM_MIN+0.44	4Ch	VCOM_MIN+0.76	6Ch	VCOM_MIN+1.08
0Dh	VCOM_MIN+0.13	2Dh	VCOM_MIN+0.45	4Dh	VCOM_MIN+0.77	6Dh	VCOM_MIN+1.09
0Eh	VCOM_MIN+0.14	2Eh	VCOM_MIN+0.46	4Eh	VCOM_MIN+0.78	6Eh	VCOM_MIN+1.10
0Fh	VCOM_MIN+0.15	2Fh	VCOM_MIN+0.47	4Fh	VCOM_MIN+0.79	6Fh	VCOM_MIN+1.11
10h	VCOM_MIN+0.16	30h	VCOM_MIN+0.48	50h	VCOM_MIN+0.80	70h	VCOM_MIN+1.12
11h	VCOM_MIN+0.17	31h	VCOM_MIN+0.49	51h	VCOM_MIN+0.81	71h	VCOM_MIN+1.13
12h	VCOM_MIN+0.18	32h	VCOM_MIN+0.50	52h	VCOM_MIN+0.82	72h	VCOM_MIN+1.14
13h	VCOM_MIN+0.19	33h	VCOM_MIN+0.51	53h	VCOM_MIN+0.83	73h	VCOM_MIN+1.15
14h	VCOM_MIN+0.20	34h	VCOM_MIN+0.52	54h	VCOM_MIN+0.84	74h	VCOM_MIN+1.16
15h	VCOM_MIN+0.21	35h	VCOM_MIN+0.53	55h	VCOM_MIN+0.85	75h	VCOM_MIN+1.17
16h	VCOM_MIN+0.22	36h	VCOM_MIN+0.54	56h	VCOM_MIN+0.86	76h	VCOM_MIN+1.18
17h	VCOM_MIN+0.23	37h	VCOM_MIN+0.55	57h	VCOM_MIN+0.87	77h	VCOM_MIN+1.19
18h	VCOM_MIN+0.24	38h	VCOM_MIN+0.56	58h	VCOM_MIN+0.88	78h	VCOM_MIN+1.20
19h	VCOM_MIN+0.25	39h	VCOM_MIN+0.57	59h	VCOM_MIN+0.89	79h	VCOM_MIN+1.21
1Ah	VCOM_MIN+0.26	3Ah	VCOM_MIN+0.58	5Ah	VCOM_MIN+0.90	7Ah	VCOM_MIN+1.22
1Bh	VCOM_MIN+0.27	3Bh	VCOM_MIN+0.59	5Bh	VCOM_MIN+0.91	7Bh	VCOM_MIN+1.23
1Ch	VCOM_MIN+0.28	3Ch	VCOM_MIN+0.60	5Ch	VCOM_MIN+0.92	7Ch	VCOM_MIN+1.24
1Dh	VCOM_MIN+0.29	3Dh	VCOM_MIN+0.61	5Dh	VCOM_MIN+0.93	7Dh	VCOM_MIN+1.25
1Eh	VCOM_MIN+0.30	3Eh	VCOM_MIN+0.62	5Eh	VCOM_MIN+0.94	7Eh	VCOM_MIN+1.26
1Fh	VCOM_MIN+0.31	3Fh	VCOM_MIN+0.63	5Fh	VCOM_MIN+0.95	7Fh	VCOM_MIN+1.27

Set Minimum VCOM Voltage Register – Address 1Ch [4:0]

DAC Value	VCOM_MIN Voltage (V)	DAC Value	VCOM_MIN Voltage (V)	DAC Value	VCOM_MIN Voltage (V)	DAC Value	VCOM_MIN Voltage (V)
00h	-3.6	08h	-2.7	10h	-1.5	18h	-0.3
01h	-3.6	09h	-2.55	11h	-1.35	19h	-0.15
02h	-3.6	0Ah	-2.4	12h	-1.2	1Ah	0
03h	-3.45	0Bh	-2.25	13h	-1.05	1Bh	0.15
04h	-3.3	0Ch	-2.1	14h	-0.9	1Ch	0.3
05h	-3.15	0Dh	-1.95	15h	-0.75	1Dh	0.45
06h	-3.0	0Eh	-1.8	16h	-0.6	1Eh	0.6
07h	-2.85	0Fh	-1.65	17h	-0.45	1Fh	0.75

VCOM Delay Time Register – Address 1Ah [4:0]

DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)	DAC Value	Delay Time (ms)
0H	0	08h	40	10h	80	18h	120
1H	5	09h	45	11h	85	19h	125
2H	10	0Ah	50	12h	90	1Ah	130
3H	15	0Bh	55	13h	95	1Bh	135
4H	20	0Ch	60	14h	100	1Ch	140
5H	25	0Dh	65	15h	105	1Dh	145
6H	30	0Eh	70	16h	110	1Eh	150
7H	35	0Fh	75	17h	115	1Fh	155

VCOM Power off Selection Register – Address 1Ah [5]

DAC Value	Power off selection	DAC Value	Power off selection
0h	VIN_UVLO_F	1h	RESET

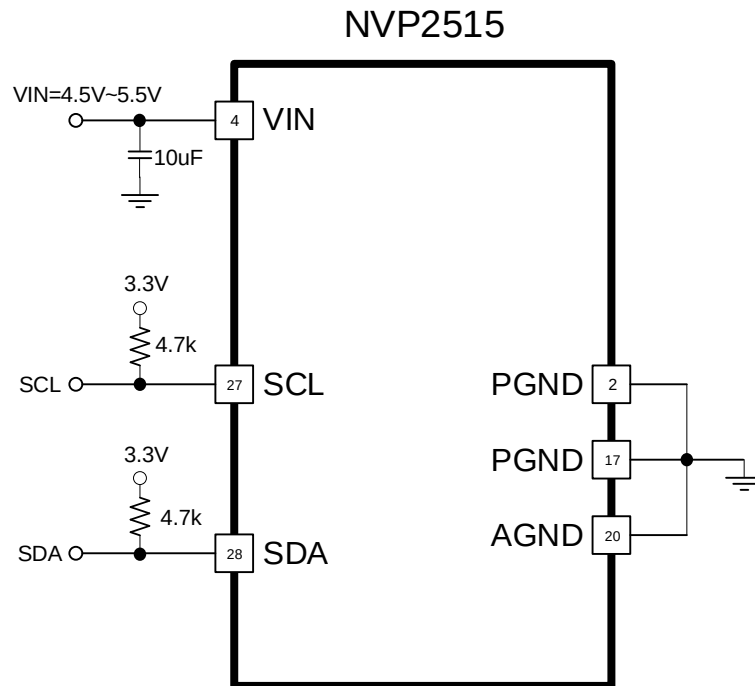
GMA1 Voltage Register – Address 0Ch [5:0]

DAC Value	GMA1 Voltage (V)	DAC Value	GMA1 Voltage (V)	DAC Value	GMA1 Voltage (V)	DAC Value	GMA1 Voltage (V)
00h	AVDD	10h	AVDD-0.32	20h	AVDD-0.64	30h	AVDD-0.96
01h	AVDD-0.02	11h	AVDD-0.34	21h	AVDD-0.66	31h	AVDD-0.98
02h	AVDD-0.04	12h	AVDD-0.36	22h	AVDD-0.68	32h	AVDD-1
03h	AVDD-0.06	13h	AVDD-0.38	23h	AVDD-0.70	33h	AVDD-1.02
04h	AVDD-0.08	14h	AVDD-0.40	24h	AVDD-0.72	34h	AVDD-1.04
05h	AVDD-0.10	15h	AVDD-0.42	25h	AVDD-0.74	35h	AVDD-1.06
06h	AVDD-0.12	16h	AVDD-0.44	26h	AVDD-0.76	36h	AVDD-1.08
07h	AVDD-0.14	17h	AVDD-0.46	27h	AVDD-0.78	37h	AVDD-1.10
08h	AVDD-0.16	18h	AVDD-0.48	28h	AVDD-0.80	38h	AVDD-1.12
09h	AVDD-0.18	19h	AVDD-0.50	29h	AVDD-0.82	39h	AVDD-1.14
0Ah	AVDD-0.20	1Ah	AVDD-0.52	2Ah	AVDD-0.84	3Ah	AVDD-1.16
0Bh	AVDD-0.22	1Bh	AVDD-0.54	2Bh	AVDD-0.86	3Bh	AVDD-1.18
0Ch	AVDD-0.24	1Ch	AVDD-0.56	2Ch	AVDD-0.88	3Ch	AVDD-1.20
0Dh	AVDD-0.26	1Dh	AVDD-0.58	2Dh	AVDD-0.90	3Dh	AVDD-1.22
0Eh	AVDD-0.28	1Eh	AVDD-0.60	2Eh	AVDD-0.92	3Eh	AVDD-1.24
0Fh	AVDD-0.30	1Fh	AVDD-0.62	2Fh	AVDD-0.94	3Fh	AVDD-1.26

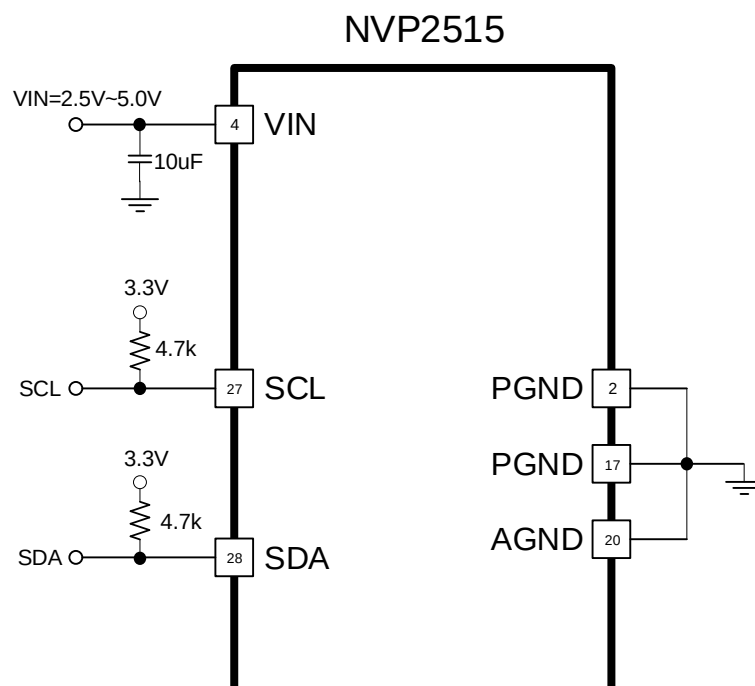
GMA2 Voltage Register – Address 0Dh [5:0]

DAC Value	GMA1 Voltage (V)	DAC Value	GMA1 Voltage (V)	DAC Value	GMA1 Voltage (V)	DAC Value	GMA1 Voltage (V)
00h	AVEE	10h	AVEE+0.32	20h	AVEE+0.64	30h	AVEE+0.96
01h	AVEE+0.02	11h	AVEE+0.34	21h	AVEE+0.66	31h	AVEE+0.98
02h	AVEE+0.04	12h	AVEE+0.36	22h	AVEE+0.68	32h	AVEE+1
03h	AVEE+0.06	13h	AVEE+0.38	23h	AVEE+0.70	33h	AVEE+1.02
04h	AVEE+0.08	14h	AVEE+0.40	24h	AVEE+0.72	34h	AVEE+1.04
05h	AVEE+0.10	15h	AVEE+0.42	25h	AVEE+0.74	35h	AVEE+1.06
06h	AVEE+0.12	16h	AVEE+0.44	26h	AVEE+0.76	36h	AVEE+1.08
07h	AVEE+0.14	17h	AVEE+0.46	27h	AVEE+0.78	37h	AVEE+1.10
08h	AVEE+0.16	18h	AVEE+0.48	28h	AVEE+0.80	38h	AVEE+1.12
09h	AVEE+0.18	19h	AVEE+0.50	29h	AVEE+0.82	39h	AVEE+1.14
0Ah	AVEE+0.20	1Ah	AVEE+0.52	2Ah	AVEE+0.84	3Ah	AVEE+1.16
0Bh	AVEE+0.22	1Bh	AVEE+0.54	2Bh	AVEE+0.86	3Bh	AVEE+1.18
0Ch	AVEE+0.24	1Ch	AVEE+0.56	2Ch	AVEE+0.88	3Ch	AVEE+1.20
0Dh	AVEE+0.26	1Dh	AVEE+0.58	2Dh	AVEE+0.90	3Dh	AVEE+1.22
0Eh	AVEE+0.28	1Eh	AVEE+0.60	2Eh	AVEE+0.92	3Eh	AVEE+1.24
0Fh	AVEE+0.30	1Fh	AVEE+0.62	2Fh	AVEE+0.94	3Fh	AVEE+1.26

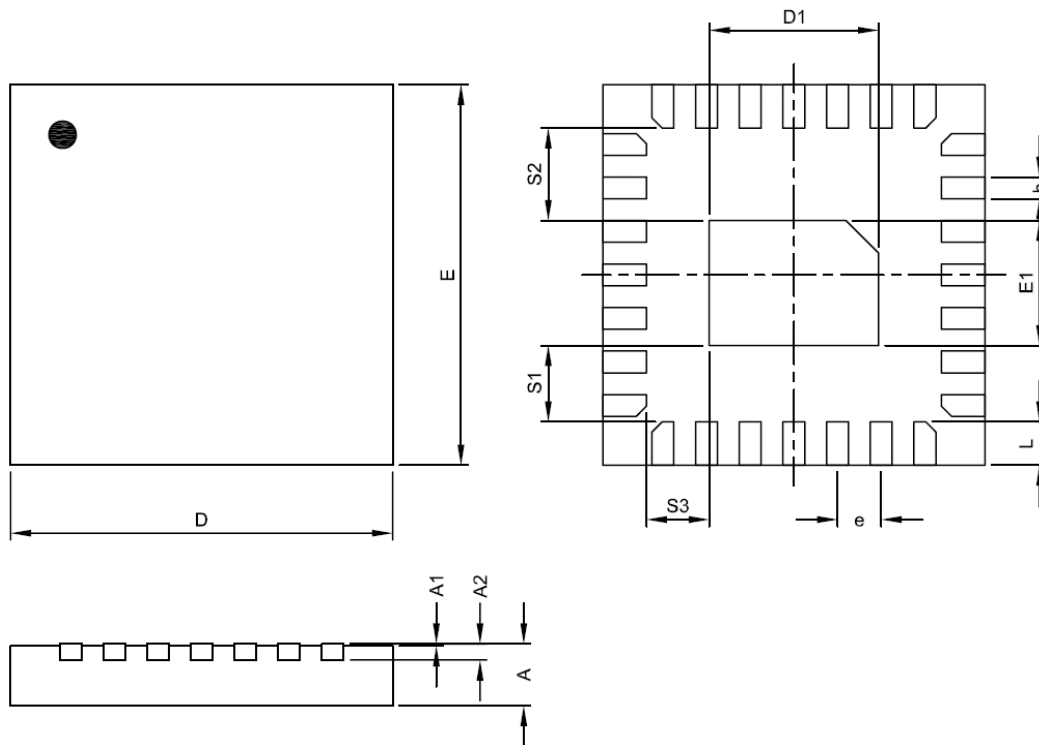
Coding Circuit for Single Chip



Coding Circuit for Panel Module



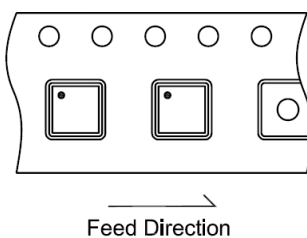
Package Information



WQFN3.5X3.5-28 Package

Symbol	DIMENSION IN MM			DIMENSION IN INCH		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.50	0.55	0.60	0.0197	0.0217	0.0236
A1	0.00	---	0.05	0.0000	---	0.0020
A2	0.150 REF			0.0059 REF		
D	3.45	3.50	3.55	0.1358	0.1378	0.1398
E	3.45	3.50	3.55	0.1358	0.1378	0.1398
D1	1.50	1.55	1.60	0.0590	0.0610	0.0630
E1	1.10	1.15	1.20	0.0433	0.0453	0.0472
b	0.15	0.20	0.25	0.0059	0.0079	0.0098
e	0.40 BSC			0.0157 BSC		
L	0.33	0.40	0.45	0.0130	0.0157	0.0177
S1	0.70 REF			0.0276 REF		
S2	0.85 REF			0.0335 REF		
S3	0.575 REF			0.0226 REF		

Taping Specification



PACKAGE	Q'TY/REEL
WQFN3.5X3.5-28	3,000 ea

NewVision does not assume any responsibility for use of any circuitry described, no circuit patent licenses are implied and NewVision reserves the right at any time without notice to change said circuitry and specifications.